

Alexander Prospal

Cleveland, OH

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Education

Bachelor of Science in Computer Science (Math Minor) Expected Graduation: May 2028

Case Western Reserve University, Cleveland, OH

Bachelor of Arts in Economics (Quantitative Methods Track) Expected Graduation: May 2028

Case Western Reserve University, Cleveland, OH

Projects

SpartieSwap — Campus Borrowing & Lending Platform CSDS 393: Software Engineering

- Built a campus-only borrowing and lending web application with a four-person Scrum team using Django, Python, PostgreSQL, Bootstrap, and Docker
- Developed a complete exchange workflow supporting item listings, search and filtering, borrowing requests, owner approval, pickup and return tracking, borrowing histories, overdue indicators, and two-way reviews
- Applied server-side authorization and relational business rules to enforce listing ownership, prevent conflicting approved loans, and protect role-specific application actions
- Integrated GitHub Actions continuous integration with PostgreSQL, Ruff linting, migration validation, and automated testing; final release completed all 10 user stories and 40 story points with 153 passing tests and 98% measured code coverage

Secure Password Manager Application CSDS 344: Computer Security

- Designed and implemented a secure password manager in Java using MVC architecture, encrypted persistent storage, and controlled credential access
- Built a JavaFX graphical interface supporting authentication and credential creation, retrieval, modification, and deletion
- Applied input validation, separation of concerns, and secure data-handling practices to improve application security and maintainability

Theater Inventory Database System CSDS 341: Introduction to Database Systems

- Designed and implemented a normalized PostgreSQL database for managing costumes, props, performers, inventory availability, and show assignments
- Defined relational schemas with primary, foreign, and composite keys and wrote SQL queries using joins, aggregation, nested subqueries, and constraint-based filtering
- Developed a Python command-line application using parameterized queries and application-level business logic for inventory tracking, availability checks, and reporting

Personal Portfolio Website prospal.dev

- Designed and developed a responsive personal portfolio using HTML, CSS, and JavaScript to showcase technical projects, academic work, and professional experience
- Deployed and maintain the site through Git-based version control and GitHub Pages with an emphasis on responsive design, accessibility, and content organization

Technical Skills

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- **Languages:** Python, Java, SQL, JavaScript, R, HTML, CSS
 - **Frameworks & Libraries:** Django, JavaFX, Flask, Bootstrap, scikit-learn, Pandas, NumPy
 - **Databases & Development Tools:** PostgreSQL, Git, GitHub, GitHub Actions, Docker, Linux, Ruff
 - **Technical Areas:** Machine Learning, Algorithms, Database Design, Secure Software Development, Software Engineering, MVC/MVT Architecture, HPC/Cluster Computing, SLURM

Relevant Coursework

Current – Fall 2026: CSDS 423 Numerical Algorithms for Machine Learning, CSDS 465 Computer Vision, CSDS 490 Digital Image Processing

Completed: Introduction to Machine Learning, Algorithms, Software Engineering, Introduction to Database Systems, Computer Security, Data Visualization

Mathematics: Linear Algebra, Calculus III, Statistics, Discrete Mathematics

Certification

Developing Secure Software (LFD121) — The Linux Foundation & OpenSSF

April 2026

Credential ID: LF-lsc03650u9 — Verified Credential